

```
aman@DESKTOP-H8PMOH6: ~/xv6-labs-2025
sixfive_all: OK (1.5s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.6s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.8s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.8s)
(Old xv6.out.find_curdir failure log removed)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.8s)
(Old xv6.out.find_subdir failure log removed)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (2.3s)
(Old xv6.out.find_recursive failure log removed)
== Test exec ==
$ make qemu-gdb
exec: OK (1.8s)
(Old xv6.out.find_exec failure log removed)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (2.1s)
(Old xv6.out.find_exec_args failure log removed)
== Test exec, recursive find ==
$ make qemu-gdb
Timeout! exec, recursive find: OK (30.3s)
(Old xv6.out.test_find_sh failure log removed)
== Test time ==
time: FAIL
Cannot read time.txt
Score: 130/131
make: *** [Makefile:361: grade] Error 1
aman@DESKTOP-H8PMOH6: ~/xv6-labs-2025$
```

aman@DESKTOP-H8PMOH6: ~/xv6-labs-2025

```
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/virtio_disk.o kernel/virtio_disk.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/start.o kernel/start.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o
kernel/switch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o
kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
make[1]: Leaving directory '/home/aman/xv6-labs-2025'
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (7.8s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (1.0s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.2s)
== Test sixfive test ==
$ make qemu-gdb
sixfive_test: OK (1.5s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (1.6s)
== Test sixfive_all ==
$ make qemu-gdb
```



```
aman@DESKTOP-H8PMOH6: ~/xv6-labs-2025
aman@DESKTOP-H8PMOH6:~/xv6-labs-2025$ make grade
make clean
make[1]: Entering directory '/home/aman/xv6-labs-2025'
rm -rf *.tex *.dvi *.idx *.aux *.log *.ind *.ilg *.dSYM *.zip *.pcap \
*/*.o */*.d */*.asm */*.sym \
kernel/kernel fs.img \
mkfs/mkfs .gdbinit \
user/usys.S \
user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc
user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user/_find user/_exec user/_sixfive user/_uptime user/_memdump
make[1]: Leaving directory '/home/aman/xv6-labs-2025'
./grade-lab-util -v
make[1]: Entering directory '/home/aman/xv6-labs-2025'
riscv64-unknown-elf-gcc -g -c -o kernel/entry.o kernel/entry.S
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/kalloc.o kernel/kalloc.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/string.o kernel/string.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/main.o kernel/main.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/vm.o kernel/vm.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/proc.o kernel/proc.c
riscv64-unknown-elf-gcc -g -c -o kernel/switch.o kernel/switch.S
riscv64-unknown-elf-gcc -g -c -o kernel/trampoline.o kernel/trampoline.S
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/trap.o kernel/trap.c
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-bui
ltin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strc
hr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf
-I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/syscall.o kernel/syscall.c
```

[illegible]